An empirical example: *Rhinoclemmys* data

Daniel R. Miranda-Esquivel

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To evaluate the effect of branch length on the area(s) chosen for conservation, I conduct a Phylogenetic Diversity (PD) analysis for the *Rhinoclemmys* genus. The topology I will use corresponds to a Total Evidence analysis from Romero-Alarcon (2020), and the distribution will be modified from Le and McCord (2008).

Read the data: the distribution, and the tree

```r
## For reproductibility purposes

set.seed(121)


options(warn=-1)
library(ggplot2)
suppressMessages(library(ggtree))

suppressMessages(library(blepd))

suppressMessages(library(phytools))

#~ library(ggtree)
##
## Just in case, if you want to install ggtree.
##
### https://bioconductor.org/packages/release/bioc/html/ggtree.html
##
####
#  if (!require("BiocManager", quietly = TRUE))
#      install.packages("BiocManager")
#  BiocManager::install("ggtree")

## Version

cat("Analyses made with blepd version:",unlist(packageVersion("blepd")))
```

```
## Analyses made with blepd version: 0 3 2 2024 7 30
```

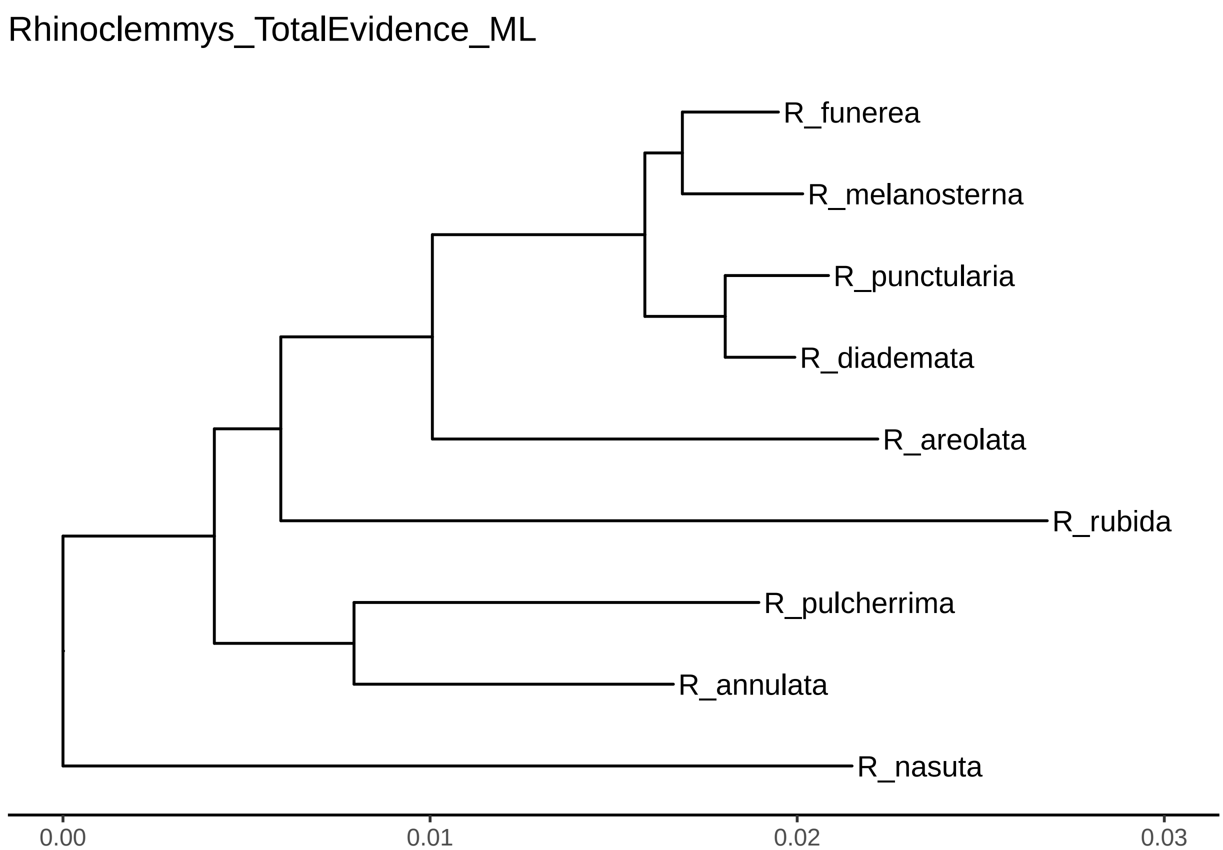
```r
## Create an object to place the distribution and the tree

RhinoclemmysData <- list()

## Read data
```

```r
## distribution is a labeled csv file, areas by terminals

#setwd("./csv/")

## getwd()

csvFile <- list.files(pattern="csv")

## csvFile

### the functions use a matrix object for the distributions

RhinoclemmysData$distribution <-  as.matrix(read.table(csvFile,
                                  stringsAsFactors=FALSE,
                                  header=TRUE,
                                  row.names=1,
                                  sep=",")
                                  )


print(t(RhinoclemmysData$distribution))
```

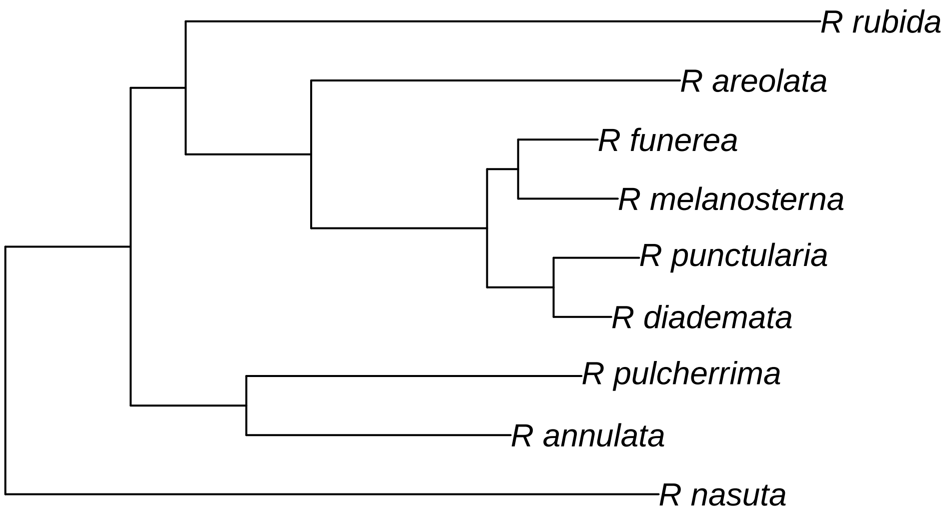
```
##                Al An-As Cho Nsa Nuh Ph Pl
## R_annulata      1     0   1   0   0  1  0
## R_areolata      1     0   0   0   0  0  0
## R_diademata     0     0   0   1   0  0  0
## R_funerea       1     0   0   0   1  0  0
## R_melanosterna  0     0   1   0   0  1  0
## R_nasuta        0     0   1   0   0  0  0
## R_pulcherrima   0     0   0   0   1  0  1
## R_punctularia   0     1   0   0   0  0  0
## R_rubida        0     0   0   0   0  0  1
```

```r
## tree(s) in nexus or newick format

##setwd("../tree/")

treeFiles <- list.files(pattern=".tre")

##treeFiles

RhinoclemmysData$tree  <-    read.tree(treeFiles)


## name of tree(s)

treeFiles <- gsub(".tre","",treeFiles)



## Plotting:

## The tree
```

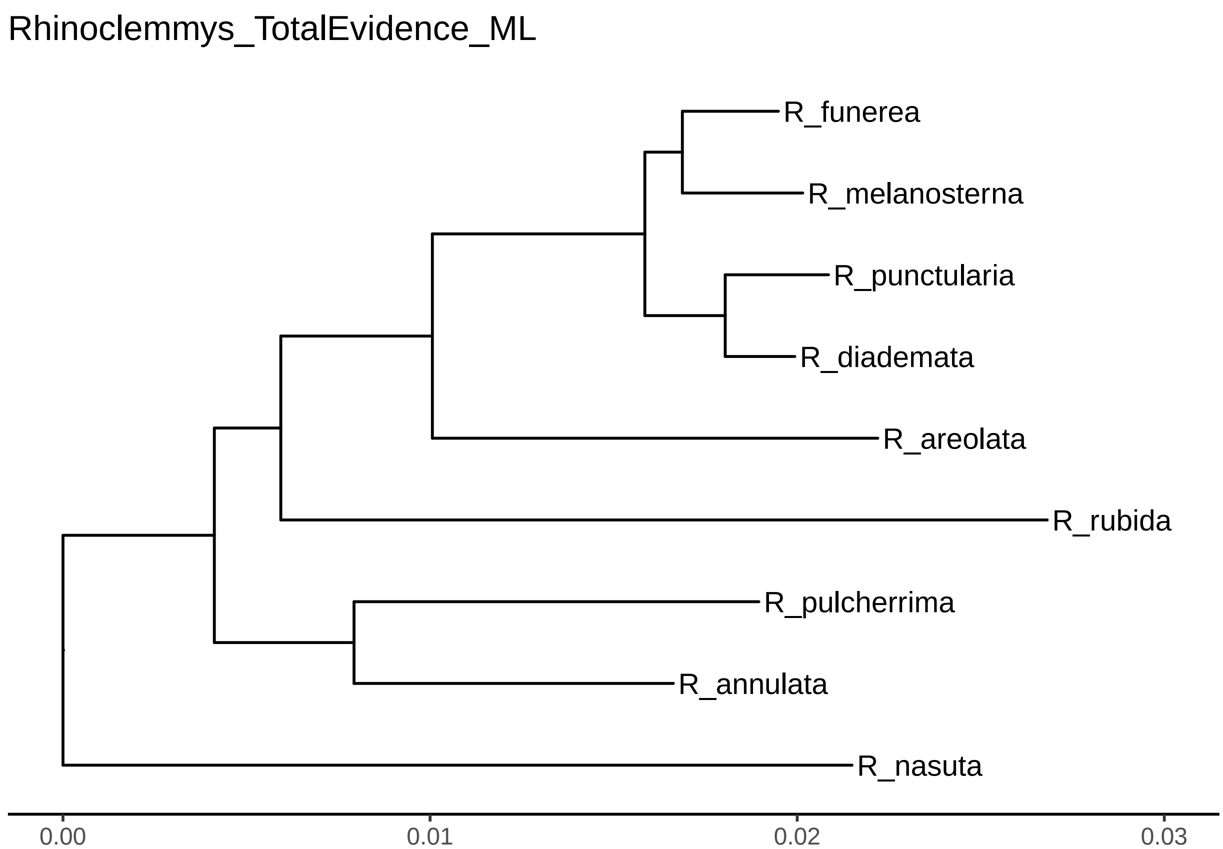
```r
## using ggtree

# RhinoclemmysData$tree <- reorder(RhinoclemmysData$tree, order = "cladewise")

plotTree <-  ggtree(RhinoclemmysData$tree, ladderize=TRUE,
                    color="black", size=0.51, linetype="solid") +
             geom_tiplab(size=4, color="black") +
             xlim(0,0.030) +
             theme_tree2() +
             ggtitle(treeFiles[1])




print(plotTree)
```




```r
#~ Alternatively, we can plot the trees using APE


plot.phylo(RhinoclemmysData$tree)
```




```
####nodelabels()
####tiplabels()

plot(plotTree)
```



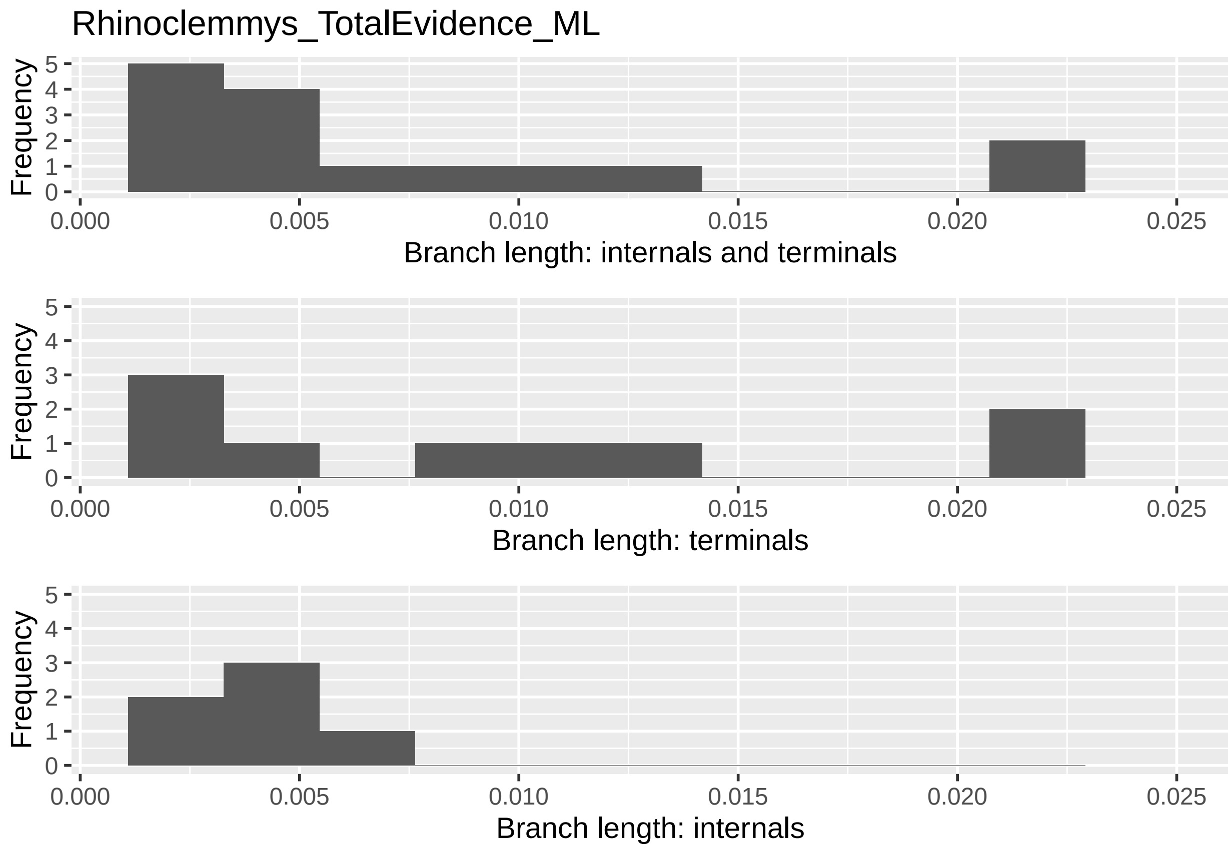

```
print(t(RhinoclemmysData$distribution))
```

```
##                Al An-As Cho Nsa Nuh Ph Pl
## R_annulata      1     0   1   0   0  1  0
## R_areolata      1     0   0   0   0  0  0
## R_diademata     0     0   0   1   0  0  0
## R_funerea       1     0   0   0   1  0  0
## R_melanosterna  0     0   1   0   0  1  0
## R_nasuta        0     0   1   0   0  0  0
## R_pulcherrima   0     0   0   0   1  0  1
## R_punctularia   0     1   0   0   0  0  0
## R_rubida        0     0   0   0   0  0  1
```

Now we plot the branch lengths

```
library(ggplot2)

##par(mfrow=c(2,1))

datos <- RhinoclemmysData$tree$edge.length
data <- as.data.frame(datos)

a <- ggplot2::ggplot(data, aes(x=datos)) +
     geom_histogram(bins = 12) +
     labs(title = treeFiles, x = "Branch length: internals and terminals", y = "Frequency") +
```

```
    xlim(c(0.001, 0.025)) +
    ylim(c(0, 5))

terminals <- RhinoclemmysData$tree$edge[,2] < 1 + length(RhinoclemmysData$tree$tip.label)

datos <- RhinoclemmysData$tree$edge.length[terminals]
data <- as.data.frame(datos)

b <- ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(x = "Branch length: terminals", y = "Frequency") +
  xlim(c(0.001, 0.025)) +
  ylim(c(0, 5))



datos <- RhinoclemmysData$tree$edge.length[!terminals]
data <- as.data.frame(datos)

c <- ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(x = "Branch length: internals", y = "Frequency") +
  xlim(c(0.001, 0.025)) +
  ylim(c(0, 5))




cowplot::plot_grid(a, b, c, nrow=3)
```




```
#
#  cat("Terminals, with BL larger than 0.02 = ",
# findTerminalgivenLength(RhinoclemmysData$tree,0.02))
#
```

The branch length histograms and the tree plot show that the internal length branches are similar among each other, and different to terminals'; there are two longer branches (larger than 0.02), *R. nasuta* (inhabiting Cho) and *R. rubida* (Pl), whereas Al and Cho are the richest.

PD values

First, I calculate the PD value for the areas, given this tree.

```
RhinoclemmysData$tablePD <- PDindex(RhinoclemmysData$tree,
                              distribution=RhinoclemmysData$distribution,
                                    root=TRUE,
                                    percentual = TRUE)

RhinoclemmysData$matrixPD <- as.data.frame(
                                  matrix(
                                      unlist(RhinoclemmysData$tablePD),
                                      nrow= length(treeFiles),
                                      byrow=TRUE))


colnames(RhinoclemmysData$matrixPD) <- row.names(RhinoclemmysData$distribution)
```

```r
row.names(RhinoclemmysData$matrixPD) <- "PD value"

 new <- (t(RhinoclemmysData$matrixPD))


values <-  new[order(new[,1])]
names  <-  rownames(new)[order(new[,1])]

print(as.data.frame(paste(names,values)))
```

```
##   paste(names, values)
## 1             Nsa 8.05
## 2           An-As 8.42
## 3             Ph 13.18
## 4            Nuh 13.86
## 5             Pl 16.81
## 6             Al 17.82
## 7            Cho 21.86
```

The highest PD is for area Cho, followed by Al, the two richest areas, and the difference in PD value is not given by the richness but the species in each region.

#Effect of branch lengths in the PD

We test the effect of the branch length on the PD values by swapping terminal and/or internal branch lengths using the three available models.

```r
 for( modelo in c("simpleswap","allswap","uniform") ){

        for( rama in c("terminals","internals","all") ){

            val <- swapBL(RhinoclemmysData$tree,
                          RhinoclemmysData$distribution,
                          model = modelo,
                          branch = rama
               )

         cat( "\n\t Tree=",treeFiles,"\n\t Model=",modelo,
             "\n\t Branchs swapped=",rama,"\n" )

        printSwapBL(val)

         cat( "\n\n" )

     }

  }
```

```
## model to test simpleswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= simpleswap
##   Branchs swapped= terminals
##   Initial    Branch      Model Al Cho Pl
## 1     Cho terminals simpleswap 13  85  2
##
```

```
##
## model to test simpleswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= simpleswap
##   Branchs swapped= internals
##   Initial    Branch      Model Cho
## 1     Cho internals simpleswap 100
##
##
## model to test simpleswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= simpleswap
##   Branchs swapped= all
##   Initial Branch      Model Al Cho Pl
## 1     Cho    all simpleswap  8  91  1
##
##
## model to test allswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= allswap
##   Branchs swapped= terminals
##   Initial    Branch   Model Al Cho Nuh Pl
## 1     Cho terminals allswap 35  53  10  2
##
##
## model to test allswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= allswap
##   Branchs swapped= internals
##   Initial    Branch   Model Cho
## 1     Cho internals allswap 100
##
##
## model to test allswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= allswap
##   Branchs swapped= all
##   Initial Branch   Model Al An-As Cho Nsa Nuh Pl
## 1     Cho    all allswap 37     4  33   4  19  3
##
##
## model to test uniform reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= uniform
##   Branchs swapped= terminals
##   Initial    Branch   Model Al Cho Nuh
## 1     Cho terminals uniform 46  50   4
##
```

```
##
## model to test uniform reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= uniform
##   Branchs swapped= internals
##   Initial    Branch   Model Cho
## 1     Cho internals uniform 100
##
##
## model to test uniform reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= uniform
##   Branchs swapped= all
##   Initial Branch   Model Al An-As Cho Nsa Nuh
## 1     Cho    all uniform 38     1  53   1   7
```

The terminal branch length is critical in the decision taken, if we swap all terminal branch lengths (model= "allswap"), or if we replace them with a uniform distribution (model= "uniform"), the area selected change from Cho to Al.

As the terminal branch lengths are distributed unequally, we might suspect that the results could depend on the longest branches that inhabit Cho/Al areas.

First, we tested whether the number of replicates had any effect.

```
for(repetir in 1:2){

            val <- swapBL( RhinoclemmysData$tree,
                           RhinoclemmysData$distribution,
                           model = "allswap",
                           branch = "terminals",
                           nTimes = 10**repetir
                         )

            printSwapBL(val)

}
```

```
## model to test allswap reps 10
##   Initial    Branch   Model Al Cho Nuh
## 1     Cho terminals allswap 10  60  30
## model to test allswap reps 100
##   Initial    Branch   Model Al Cho Nsa Nuh Pl
## 1     Cho terminals allswap 39  40   1  19  1
```

Roughly speaking, from 1000 on the results are alike, and the largest The difference was the use of 10 or 100 replicates. As a rule of thumb, we must use at least 100 replicates, but 1000 replicates would be better.

Now, let us see if the difference in results could be assigned to the longest branches (or not).

```
## we can use a single command (*evalBranch*)

AllTerminals <- evalBranch(tree=RhinoclemmysData$tree,
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="terminals",
                           approach="all",
```

```r
                             printNames=TRUE)

printEvalBranch(AllTerminals)
```

```
##                  node initialArea FinalArea Aproach   %Delta
## 1       [R_nasuta]|1         Cho        Al   lower  -45.859
## 12 [R_pulcherrima]|3         Cho        Pl   upper  115.809
## 13   [R_diademata]|4         Cho       Nsa   upper 1710.405
## 14 [R_punctularia]|5         Cho     An-As   upper 1103.961
## 16     [R_funerea]|7         Cho        Al   upper  325.502
## 17    [R_areolata]|8         Cho        Al   upper   86.219
## 18      [R_rubida]|9         Cho        Pl   upper   59.024
```

```r
AllInternals <- evalBranch(tree=RhinoclemmysData$tree,
                             distribution=RhinoclemmysData$distribution,
                             branchToEval="internals",
                             approach="all",
                             printNames=TRUE)

printEvalBranch(AllInternals)
```

```
##                                              node initialArea FinalArea Aproach
## 13 [node number:16:R_diademata/R_punctularia]|16         Cho     An-As   upper
##      %Delta
## 13 1694.723
```

The selected area depended on the branch length. While the internal terminals had almost no impact, if the terminal branch length of *R. nasuta* is -47.18% shorter, or if the terminal branch length is 84.95% larger for *R. areolata*, the area selected will change from Cho to Al, and if the terminal branch length of *R. rubida* is 61.32% larger, and the area selected changes from Cho to Pl.

If the tree is ultrametric, the pair of sister pairs will behave alike, and the difference will be given by the distribution.

```r
## now let's see the impact of ultra-metricity

AllTerminalsU <- evalBranch(tree=force.ultrametric(RhinoclemmysData$tree),
                             distribution=RhinoclemmysData$distribution,
                             branchToEval="terminals",
                             approach="all",
                             printNames=TRUE)
```

```
## ***************************************************************
## *                          Note:                              *
## *    force.ultrametric does not include a formal method to    *
## *    ultrametricize a tree & should only be used to coerce    *
## *   a phylogeny that fails is.ultrametric due to rounding --  *
## *    not as a substitute for formal rate-smoothing methods.   *
## ***************************************************************
```

```r
cat("Ultrametric tree\n")
```

```
## Ultrametric tree
```

```r
printEvalBranch(AllTerminalsU)
```

```
##                  node initialArea FinalArea Aproach   %Delta
## 9       [R_nasuta]|1         Cho        Al   lower  -48.953
```

```
## 11      [R_funerea]|7         Cho         Al   upper  336.352
## 12    [R_diademata]|4         Cho        Nsa   upper 1890.607
## 13 [R_punctularia]|5         Cho      An-As   upper 1890.607
## 14     [R_areolata]|8         Cho         Al   upper   92.029
## 15       [R_rubida]|9         Cho         Pl   upper  134.869
## 17 [R_pulcherrima]|3         Cho      NuhPl   upper  209.954
```

```r
printEvalBranch(AllTerminals)
```

```
##                  node initialArea FinalArea Aproach   %Delta
## 1        [R_nasuta]|1         Cho        Al   lower  -45.859
## 12 [R_pulcherrima]|3         Cho        Pl   upper  115.809
## 13    [R_diademata]|4         Cho       Nsa   upper 1710.405
## 14 [R_punctularia]|5         Cho     An-As   upper 1103.961
## 16      [R_funerea]|7         Cho        Al   upper  325.502
## 17     [R_areolata]|8         Cho        Al   upper   86.219
## 18       [R_rubida]|9         Cho        Pl   upper   59.024
```

```r
AllInternalsU <- evalBranch(tree=force.ultrametric(RhinoclemmysData$tree),
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="internals",
                           approach="all",
                           redondeo=3,
                           printNames=TRUE)
```

```
## ***************************************************************
## *                          Note:                              *
## *    force.ultrametric does not include a formal method to    *
## *    ultrametricize a tree & should only be used to coerce    *
## *   a phylogeny that fails is.ultrametric due to rounding --  *
## *    not as a substitute for formal rate-smoothing methods.   *
## ***************************************************************
```

```r
printEvalBranch(AllInternalsU)
```

```
##                                              node initialArea FinalArea Aproach
## 8 [node number:16:R_diademata/R_punctularia]|16         Cho  An-AsNsa   upper
##    %Delta
## 8 1860.13
```

```r
printEvalBranch(AllInternals)
```

```
##                                               node initialArea FinalArea Aproach
## 13 [node number:16:R_diademata/R_punctularia]|16         Cho     An-As   upper
##      %Delta
## 13 1694.723
```

Literature cited

Le, Minh, and William P. McCord. 2008. "Phylogenetic relationships and biogeographical history of the genus Rhinoclemmys Fitzinger, 1835 and the monophyly of the turtle family Geoemydidae (Testudines: Testudinoidea)." *Zoological Journal of the Linnean Society* 153 (4): 751–67. https://doi.org/10.1111/j.1096-3642.2008.00413.x.

Romero-Alarcon, L. Viviana. 2020. "A total-evidence phylogeny of the crown and stem-groups of turtles (Pantestudines: Testudinata)." Master's thesis, Colombia: Escuela de Biología, UIS.